

Distributed Unsupervised Learning for Combinatorial User Assignment in mmWave Cell-Free Massive MIMO Using Graph Neural Networks

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Abstract—Smaller cells have been the most important contributor to throughput improvement since the birth of cellular networks. They are likely to evolve further in the shift to cell-free massive MIMO (CF mMIMO), where multiple closely placed access points (APs) collaborate to serve users. This scheme is particularly suitable for millimeter wave (mmWave) communication,

which enables very high data rates with its large bandwidth, but encounters severe challenges of high path loss and blockage. The CF mMIMO network is a good countermeasure to these two challenges by utilizing overlapping signals from different APs and macro-diversity. In this work, we demonstrate that mmWave CF mMIMO network optimization is largely an AP-user assignment problem. To solve this large-scale, nondifferentiable problem, we propose an unsupervised machine learning (ML) approach, which looks for the optimal solution autonomously without labels. A customized graph neural network architecture tailored to the problem properties is proposed, which enables distributed optimization without a central unit, allows for a varying number of users, and hierarchical permutation-equivariance of APs and users. A teacher-student model is applied to prune the graph, where the teacher model uses a fully connected graph for maximum performance, and the student model uses a pruned graph to reproduce the teacher's behavior with less communication in fronthaul. Moreover, a special training method is designed, which relaxes the combinatorial problem to a continuous one. In this way, we can apply gradient-based neural network training. An entropy-inspired penalty is introduced to make the relaxed problem equivalent to the original one. The analytical augmented Lagrangian method is combined with ML for the constrained optimization. Simulation results show that the proposed approach outperforms baselines in both performance and computation time. In addition, with a properly pruned graph, the proposed approach performs inference in a distributed manner with sparse message passing between APs, realizing a low signaling overhead in fronthaul, and a performance close to the fully connected graph.

Index Terms—Cell-free massive MIMO, combinatorial optimization, graph neural network, graph pruning, permutation-equivariance, machine learning, teacher-student model.

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I. INTRODUCTION

WE HAVE witnessed a massive increase in wireless communication capacity by a factor of a million over the past few decades. Among all contributing factors, the most important one is the reduced transmit distance with a contribution of 1600-fold gain, while all other factors combined contributed a 625-fold gain [2]. As the distance between base stations (BSs) decreases, BSs are required to coordinate

with each other [3]. This trend is likely to lead to a shift to cell-free massive MIMO (CF mMIMO), where densely deployed access points (APs) collaborate to serve users [4]. In recent years, CF mMIMO has attracted interest from both academia and industry for various reasons. The collaboration of APs allows for a higher received signal strength (RSS) due to the overlapping received signals from different APs. It also realizes macro-diversity because signals come from different directions and it is unlikely that they are blocked simultaneously. These advantages make CF mMIMO suitable for millimeter wave (mmWave) communication. Despite the high data rate enabled by large bandwidth, mmWave communication is limited by high path loss and high blockage loss, which can be compensated by signal overlapping and macro-diversity, respectively.

The hardware and channel properties of mmWave communication are very different from conventional communication systems. MmWave channels exhibit high path loss and predominantly specular propagation paths (i.e., signals travel through only a few strong and directional paths rather than scatter uniformly from all directions). To overcome high path loss, high-gain antennas with strong directivity are essential, resulting in significantly reduced interference between users compared to lower-frequency systems. Thereby, unlike communication at lower frequencies relying on sophisticated precoding techniques, such as zero-forcing (ZF) and minimum mean square error (MMSE) [5], simple conjugate precoding would be sufficient for mmWave communication. Additionally, while the theoretically optimal transmit power at mmWave is high due to the high path loss, the actual achievable transmit power is constrained by limited output power of mmWave power amplifiers. Therefore, explicit power control algorithms become unnecessary, because the maximum available amplifier output power is almost always the optimal feasible solution. As a result, mmWave CF mMIMO network optimization is largely an assignment problem between APs and users, because conjugate precoding is determined given the assignment and the transmit power is fixed.

In the literature, various approaches have been proposed to optimize resource allocation, including AP-user assignment, in CF mMIMO networks. A joint optimization of beamforming and resource allocation is proposed for mmWave CF mMIMO network in [6]. Analog beam selection and digital beamforming are jointly optimized in [7]. A mixed-integer programming formulation for resource allocation in fronthaul is presented in [8]. Energy efficiency in CF mMIMO is optimized with consideration of fronthaul capacity for both coherent and non-coherent transmission in [9]. Radio head selection and beamforming design are jointly optimized considering fronthaul capacity in [10]. A distributed and rule-based method for assignment of APs and pilots is proposed in [11]. A hybrid beamforming approach is proposed considering wide-band channel properties in [12]. An alternative selection approach on the assignment between APs and users is proposed in [13]. A greedy AP-user assignment algorithm is proposed in [14]. A learning-based method is combined with successive convex approximation (SCA) for AP-user assignment in [15]. Beamforming and AP selection are jointly optimized with

the alternating average reflection algorithm (AARA) in [16]. In [17], a joint optimization strategy employing the block-sparsity norm $l_{2,1}$ is introduced to optimize both AP-user assignment and power control simultaneously. A two-stage training is formulated for user assignment and beamforming design in [18]. Tabu search is applied for user assignment in [19]. Pilot assignment methods leveraging serving-set-based schemes and graph coloring techniques are studied in [20] and [21], respectively.

We can conclude from the references listed above that the assignment problem in the optimization of CF mMIMO is challenging due to its nondifferentiability. Existing algorithms are either suboptimally heuristic or lack scalability. In recent years, machine learning (ML) has emerged as a promising approach [22], in particular, for combinatorial problems [23], by training neural networks (NNs) on large datasets to learn optimized mappings from problems to solutions. In this framework, an encoder-decoder-style NN is employed in [24] to optimize AP scheduling using supervised learning. In [25], a “meta-algorithm” is introduced that combines reinforcement learning (RL) with graph embedding to generate greedy policies for combinatorial problems. The power allocation is optimized considering pilot contamination in [26]. RL is also used in [27] to optimize pilot assignment in a decentralized manner.

Among various ML approaches for CF mMIMO optimization, the graph neural network (GNN) architecture has attracted particular interest due to its distributed nature and high scalability. The inference of GNN is performed on a graph of nodes and edges between them. Since vertices and edges share the same information processing, the GNN realizes a high scalability [28]. In the literature, AP selection is performed with inductive graph learning in [29]. Power allocation is performed with a heterogeneous graph in [30]. The nonconvex problem of energy efficiency maximization is addressed in [31] and [32]. The minimum signal-to-interference-plus-noise ratio (SINR) is maximized with a GNN in [33]. The sum-rate is maximized with GNN and multi-agent reinforcement learning (MARL) in [34]. Despite their considerable potential, combinatorial assignment problems involving connection number constraints remain, to the best of our knowledge, an open challenge.

Although the distributed nature of GNNs enables a fully decentralized optimization of CF mMIMO network [28], [29], [30], [32], [33], the message passing between nodes (i.e., APs) is realized as traffic in the fronthaul connecting APs, which is an important consideration in CF mMIMO network optimization [8]. However, most existing works on GNNs for CF mMIMO assume a fully connected graph, resulting in heavy and unnecessary traffic in fronthaul. In a broader sense, a sparse NN is obtained with the L_0 norm as a penalty during training [35]. Inference speed is significantly improved with network pruning in [36]. Network pruning is also proposed for federated learning in [37]. A teacher-student model is proposed for large language models with sparsity [38].

When “correct” labels are not available and NNs have to look for optimal solutions autonomously (many complex problems are this case), a severe challenge is the scalability

due to the “curse of dimensionality” [39]. There are multiple ML based approaches for combinatorial problems. We can list the probabilities of all possible solutions as output of the NN, and optimize the categorical probability distribution [40]. However, this approach is not scalable because the possible solutions (the combinations of all possible assignments rather than the assignments themselves) have to be explicitly listed. We can also round the continuous NN output to discrete values [41]. This approach is constrained because we cannot apply rounding in training, since rounding is nondifferentiable and the gradient is necessary for training. If we only round the NN output in testing, an inconsistency between training and testing might arise. A third possibility is the CONTinuous relaxation for disCRETE (CONCRETE) distribution, which relaxes the discrete optimization variables to continuous ones with a hyperparameter *temperature* [42]. In this approach, tuning the temperature might be a difficulty due to the high sensitivity of performance to the temperature.

Summarizing the literature review above, we realize that a sophisticated solution to the AP-user assignment problem in mmWave CF mMIMO networks is still unavailable. For one thing, the nondifferentiable combinatorial nature of the assignment problem makes rigorous analytical solutions very difficult and heuristic methods suboptimal. For another, the specific problem properties (e.g., distributed nature of CF mMIMO networks, varying number of users, and hierarchical permutation-equivariance (HPE), as will be explained in Section II-B) are not reflected in conventional ML approaches. In particular, the nondifferentiable combinatorial problem cannot be optimized with gradient-based methods, which are the canonical NN training approach.

In this work, we propose an unsupervised ML approach for the assignment problem. A highly customized GNN is tailored according to the problem properties. The teacher-student model is applied to learn a sparse graph in order to minimize communication overhead in the fronthaul. A relaxation approach with a penalty term inspired by information entropy is proposed to optimize the combinatorial problem. The analytical augmented Lagrangian method (ALM) is combined with ML as a hybrid solution. Our test result confirms advantages of the proposed approach compared to baselines in performance, computation time, distributed inference, and overhead in fronthaul. Compared to our preliminary work [1], this work makes several substantial advancements. First, it provides a formal proof of the HPE property. Second, it investigates the generalizability of the customized GNN architecture across varying numbers of users. Third, it introduces an analytical graph pruning strategy based on a teacher-student model. These contributions collectively mark a significant improvement over the earlier work.

II. PROBLEM STATEMENT

A. System Model and Problem Formulation

We consider a mmWave CF mMIMO network with distributed APs and users, as shown in Figure 1. Users in this network periodically transmit orthogonal pilot signals to APs, allowing each AP to estimate channel gain and angle-of-arrival (AoA) of each user. Every AP is equipped with

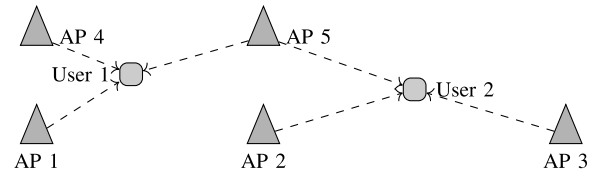


Fig. 1. The considered scenario.

a large phased array with an analog beamformer, which uses conjugate precoding to achieve high antenna gain and high directivity. The high directivity results in negligible interference between users. For example, according to the Friis transmission equation, a transmission distance of 20m at 28 GHz has a path loss of about 87 dB. As a reasonable assumption, we require an antenna gain of about 30 dBi at the AP, which requires a phased array with 256 elements. With such a high antenna gain, the effective antenna gain is about -17 dBi at an angle deviation of 10° away from the main lobe direction. The interference is hence negligible compared to the noise power even assuming an omni-directional antenna with 0 dBi gain at the user. High signal strength and negligible interference can be achieved in this way with large antenna arrays and simple algorithms. Note that APs are usually at a higher altitude than users. Even if the horizontal angle-of-arrivals of two users are close, the vertical directivity can still help separate them, because the vertical angle-of-arrivals of two users are different if their distances to the AP are different. Based on these considerations and realistic quantitative assessment that, for this type of system, multiuser interference is not the dominant effect in system optimization, we believe that our system model provides a simplified and yet realistic abstraction, which allow us to make progress in this difficult problem. Moreover, conventional transmit power control is not necessary for mmWave communications because a high path loss requires a high theoretically optimal transmit power, which is usually much higher than the maximum output power of the actual mmWave power amplifier. Therefore, the feasible optimal transmit power is almost always the maximum power. Summarizing the considerations above, the mmWave CF mMIMO network optimization problem can be formulated as an assignment problem between APs and users. Once the assignment is obtained, the conjugate precoding vector is determined by user’s AoA. Furthermore, to ensure macro-diversity as a countermeasure against blockage, which is a primary challenge in mmWave communications due to the weak diffraction effect on conventional objects at high frequency, we impose that each user must be served by at least V APs. This redundancy improves the reliability of the link in environments with dynamic obstructions [43]. Finally, due to hardware limitations, we constrain that each AP can serve at most U users. The optimization objective is to maximize the sum-rate of all users.¹ Denote the effective channel gain (the product of channel gain and antenna gain) between AP n and

¹In addition to sum-rate maximization, the proposed optimization framework can be applied for other objectives and statistical channel state information at the transmitter (CSIT) due to the generalizability of the proposed optimization framework, as will become clear in Section III.

user k without blockage as g_{kn} ,² the assignment, whether AP n serves user u , as s_{kn} ($s_{kn} = 1$ if AP n serves user k and $s_{kn} = 0$ otherwise), the problem can be formulated as

$$\max_{\mathbf{S}} \quad f = \sum_{k=1}^K \log_2 \left(1 + \frac{\sum_{n=1}^N g_{kn} s_{kn}}{\sigma^2} \right) \quad (1a)$$

$$\text{s.t.} \quad s_{kn} \in \{0, 1\}, \quad (1b)$$

$$\sum_{k=1}^K s_{kn} \leq U, \quad \forall n \quad (1c)$$

$$\sum_{n=1}^N s_{kn} \geq V, \quad \forall k \quad (1d)$$

where $\mathbf{S} \in \{0, 1\}^{K \times N}$ with s_{kn} in row k and column n , and σ^2 is the noise power. In (1), we make two reasonable assumptions of Gaussian inputs [43], [44] and noncoherent overlapping of signals from different APs because the phase synchronization between APs is very difficult at mmWave range. According to [44, Eq. 12], the lower bound on the ergodic achievable rate is given by (1a).

B. Problem Properties

Problem (1) possesses several important properties that the solution must take into consideration:

- 1) *Combinatorial nature*: The optimization variable $\mathbf{S} \in \{0, 1\}^{K \times N}$ in (1) is discrete, indicating whether an AP serves a particular user or not. This makes the problem non-differentiable and more difficult to solve than problems with continuous optimization variables.
- 2) *Scalability*: The solution must be computationally efficient and scalable with respect to the numbers of APs and users, as CF mMIMO networks typically involve a large number of both.
- 3) *Fixed number of APs, varying number of users*: In practical deployments, the number of APs is fixed, while the number of mobile users varies over time. The solution should generalize well to different numbers of users.
- 4) *Hierarchical permutation-equivariance (HPE)*: The solution should be equivariant with respect to permutations of APs and users of every AP. Specifically:
 - Permuting the indices of the APs should result in the same permutation of the output, i.e., columns of $\mathbf{S}$, since the index does not carry physical meaning and the AP order should be arbitrary.
 - Similarly, permuting user indices should also result in the same permutation of the output, i.e., rows of $\mathbf{S}$.

This ensures that the model does not depend on any specific ordering and is robust to changes in index assignment. In general, HPE can be achieved with either

training data augmentation [45], [46] or NN architecture design [47]. We prefer the latter possibility because HPE is ideally an inherent nature rather than an acquired one, and data augmentation increases resource consumption of training significantly.

- 5) *Distributed implementation with minimum overhead in fronthaul*: To accommodate the decentralized CF mMIMO network, the solution should support a distributed architecture without a central unit to control the APs. Furthermore, data traffic in the fronthaul should be minimized since signaling overhead is a major limiting factor for CF mMIMO networks with a large number of APs.

In the following sections, we introduce a highly customized GNN and a special training method that meet the above-mentioned requirements.

III. FRAMEWORK OF UNSUPERVISED MACHINE LEARNING FOR OPTIMIZATION

We propose an unsupervised ML approach to solve Problem (1). Its objective f depends on the assignment matrix $\mathbf{S}$ and channel matrix $\mathbf{G} \in \mathbb{R}^{K \times N}$, where g_{kn} is its element in row k and column n . Therefore, the objective can be written as

$$f(\mathbf{G}, \mathbf{S}). \quad (2)$$

We define an NN N_θ parameterized by θ (i.e., θ is the set of all trainable parameters of N_θ) to map from $\mathbf{G}$ to $\mathbf{S}$:

$$\mathbf{S} = N_\theta(\mathbf{G}). \quad (3)$$

Combining (2) and (3), we can write the objective as $f(\mathbf{G}, N_\theta(\mathbf{G}); \theta)$. Note that we explicitly specify that f is parameterized by θ , i.e., if we change the mapping N_θ from $\mathbf{G}$ to $\mathbf{S}$, f is changed given the same $\mathbf{G}$. Applying the data-driven optimization approach, we collect massive channel data in a training set $\mathcal{D}$, and formulate the NN training problem as

$$\max_{\theta} J = \sum_{\mathbf{G} \in \mathcal{D}} f(\mathbf{G}, N_\theta(\mathbf{G}); \theta). \quad (4)$$

If f is known and differentiable, the training can be done via *model-based learning*³. We first compute the gradient $\nabla_{\theta} J$ with respect to θ , then update NN parameter θ as

$$\theta \leftarrow \theta + \eta \nabla_{\theta} J, \quad (5)$$

where η is the learning rate. In this way, N_θ is trained to maximize f over the ensemble of $\mathbf{G} \in \mathcal{D}$. If well trained, we can expect that $\mathbf{S}' = N_\theta(\mathbf{G}')$ is also a good solution for $\mathbf{G}' \notin \mathcal{D}$. The computation of NN performance using data samples not seen in the training set is known as testing, which is essential to evaluate the generality of the trained NN on unseen data. In this way, most computationally intensive operations are shifted from deployment to training because NN N_θ only needs to be trained once and can be applied to any new channel matrix $\mathbf{G}'$ with significantly lower computational cost in inference,

²Blockage is usually sporadic in typical mobile radio scenarios and difficult to predict without knowledge of ambient environment. The channel gain without blockage can be relatively reliably estimated with pilot signals. Therefore, we use the channel gain without blockage as the input of the optimization problem. The macro-diversity is a robust counter-measure against blockage if it happens.

³The *model-free* does not require a known and differentiable objective. It uses trial-and-error to optimize the NN. However, sample efficiency and scalability of model-free learning is worse than model-based learning. In this work, we focus on the model-based learning.

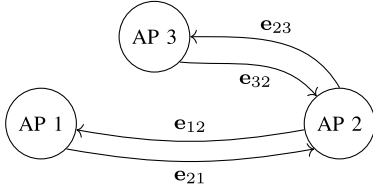


Fig. 2. An illustration of a graph. A node in the graph represents an AP. The edges between nodes enable cooperation between APs and are implemented via fronthaul links between APs.

until the network deployment changes significantly and the trained NN is no longer valid. Since the network deployment is much more expensive compared to NN training, this approach is more practical than traditional optimization algorithms that need to be executed every time when the channel changes even if the NN is trained for a specific network deployment.

The unsupervised ML described above is a general approach for many optimization problems. In the following sections, we propose a specific ML approach tailored for the considered problem to meet the requirements listed in Section II-B.

IV. CUSTOMIZED GRAPH NEURAL NETWORK FOR CF mMIMO OPTIMIZATION

A. General Graph Neural Network Architecture

The GNN architecture is chosen as N_θ according to the distributed nature of CF mMIMO. A GNN performs inference on a graph consisting of nodes and directed edges between them. A simple example of GNN is shown in Figure 2. In the graph, every node stands for an AP and has a *node feature*, which is updated by inference from layer to layer. The final node feature is the output of the GNN. Edges between APs enable cooperation between APs. Every edge has an *edge feature*, which is constant throughout the inference.⁴

Remark 1: We let every node represent an AP because nodes pass messages via edges to each other. This message passing can be implemented via fronthaul links between APs. If we introduce another type of node to represent users, the message passing between APs and users must be implemented via wireless links, which is difficult because the wireless links have not yet been established when the user assignment is decided. Furthermore, there are a certain number of APs serving a varying number of users. A fixed number of nodes in the graph cannot represent the varying number of users. The varying number of users will be processed in the specific NN architecture described in Section IV-B.

The GNN has L layers. In layer l , the inference of node features is performed as [48]

$$\mathbf{F}_n^{(l+1)} = \gamma^{(l)} \left(\mathbf{F}_n^{(l)}, \bigoplus_{m \in \mathcal{N}(n)} \phi^{(l)} \left(\mathbf{F}_m^{(l)}, \mathbf{e}_{mn} \right) \right), \quad (6)$$

where $\mathbf{F}_n^{(l+1)}$ is the node feature of node n as the output of layer l , $\gamma^{(l)}$ is an NN serving as the node processing unit in layer l , $\mathbf{F}_n^{(l)}$ is the node feature of node n as the input of layer l , $\mathcal{N}(n)$ is the set of neighboring nodes of node n

⁴In some variations of GNN, edge features can also be updated by inference. In this work, we adopt the architecture described in [48] for its simplicity and good performance.

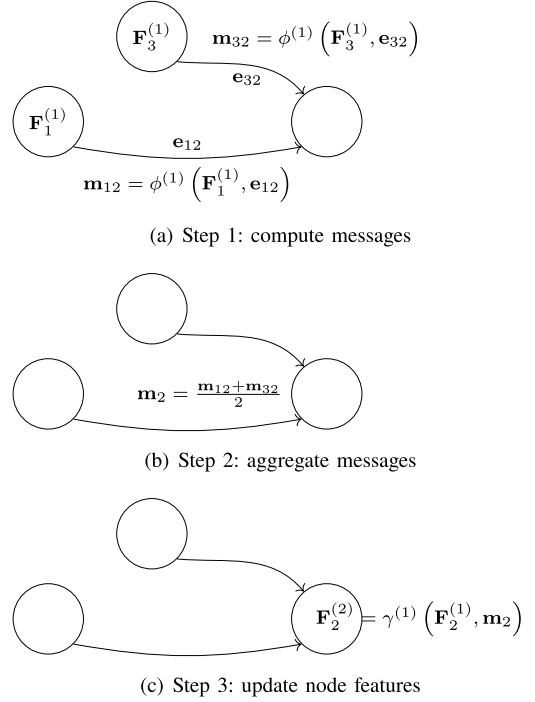


Fig. 3. Illustration of GNN inference.

(e.g., $\mathcal{N}(2) = \{1, 3\}$ in Figure 2), $\bigoplus_{m \in \mathcal{N}(n)}$ is an aggregation function over all neighboring nodes of node n , $\phi^{(l)}$ is an NN serving as the edge processing unit of layer l , $\mathbf{e}_{mn}$ is the constant edge feature of the edge from node m to node n . In this work, aggregation $\bigoplus$ is the mean of all messages from neighboring nodes. Neural networks (NNs) $\gamma^{(l)}$ and $\phi^{(l)}$ will be explained in detail in Section IV-B. Inference in layer l (6) can be interpreted as the following steps: In Step 1, every node uses $\phi^{(l)}$ to compute messages for its neighboring nodes using corresponding node feature $\mathbf{F}_m^{(l)}$ and edge feature $\mathbf{e}_{mn}$, as shown in Figure 3(a). In Step 2, all messages from a node's neighboring nodes are aggregated using the operator $\bigoplus$, as shown in Figure 3(b). In Step 3, every node uses $\gamma^{(l)}$ to generate its node feature $\mathbf{F}_n^{(l+1)}$ as output of layer l according to $\mathbf{F}_n^{(l)}$ and the aggregated message computed in Step 2, as shown in Figure 3(c). In this way, distributed inference is realized that relies on message passing between nodes rather than a central unit. Furthermore, high scalability is achieved because the same $\phi^{(l)}$ is applied to all edges and the same $\gamma^{(l)}$ is applied to all nodes.

B. Problem-Specific Graph Neural Network for CF mMIMO Optimization

The general GNN described in Section IV-A achieves scalability and distributed inference. However, it uses conventional fully connected NNs as $\phi^{(l)}$ and $\gamma^{(l)}$ [48], which cannot handle a varying number of users and does not realize HPE. In this work, we propose a highly customized GNN to fully meet the requirements specified in Section II-B.

This customized architecture is inspired by the fact that the decision on AP-user assignment depends on both channel gains of individual users and a global context shared by all

users as a reference. For layer 1, the node feature $\mathbf{F}_n^{(1)}$ of node n consists of column n of $\mathbf{G}$ and the *residual demand* of each user, defined as the number of additional APs required for each user to fulfill Constraint (1d) (the purpose of this input will become clearer later). The edge feature $\mathbf{e}_{nm}$ is determined by its origin, node n , and is independent from the destination, node m . A significant advantage of this edge definition is that input information for $\phi^{(l)}$ is available at one node (i.e., AP n). Therefore, the message computation can be done locally without additional communication in fronthaul. Using these input features, we define a *local feature* for every individual user and a *global feature* shared by all users, which are computed as

$$\mathbf{f}_{kn,j+1} = \begin{pmatrix} \text{ReLU}(\mathbf{W}_j^1 \mathbf{f}_{kn,j} + \mathbf{b}_j^1) \\ (\sum_{k'} \text{ReLU}(\mathbf{W}_j^g \mathbf{f}_{k'n,j} + \mathbf{b}_j^g)) / K \end{pmatrix}, \quad (7)$$

where $\mathbf{W}_j^1 \in \mathbb{R}^{P \times Q}$ is the weight matrix on layer l for the local feature (with superscript 1); $\mathbf{f}_{kn,j} \in \mathbb{R}^{Q \times 1}$ is the input feature of user k and AP n to layer j ; $\mathbf{b}_j^1 \in \mathbb{R}^{P \times 1}$ is the bias vector on layer j for the local feature; $\mathbf{W}_j^g$ and $\mathbf{b}_j^g$ are comparable to $\mathbf{W}_j^1$ and $\mathbf{b}_j^1$ but are used for global features shared by all users (with superscript g). The first line of (7) is the calculation of the local feature for individual users. Similarly to conventional fully connected layers, a linear transform is applied to the input feature vector, followed by the nonlinear activation function ReLU. Unlike conventional fully connected layers, where both input and output features are the complete feature, input and output features $\mathbf{f}_{k,j}$ and $\mathbf{f}_{k,j+1}$ are only for user k . The second line of (7) is the calculation of the global feature shared by all users, where the same computation as the local feature calculation is done for every user and is then averaged over all users. In this way, all users have the same global feature, which depends on the input features of all users. Finally, local and global features are aggregated in the feature dimension. Therefore, $\mathbf{f}_{kn,j+1}$ has the shape of $2P \times 1$.

In the last layer (layer J^5), we do not distinguish between local and global features such that the message passed in the fronthaul is compact. The inference of the last layer is performed as

$$\mathbf{f}_{kn,J} = \text{ReLU}(\mathbf{W}_{J-1} \mathbf{f}_{kn,J-1} + \mathbf{b}_{J-1}), \quad (8)$$

where $\mathbf{W}_{J-1}$ is the weight matrix and $\mathbf{b}_{J-1}$ is the bias vector. The information processing is intuitively illustrated in Figure 4.

A significant advantage of this customized NN architecture is that it can handle a varying number of users because the same information processing (7) and (8) is applied to all users. Compared to this arrangement, conventional fully connected layers adopted in standard GNNs [48], [49] cannot handle a varying number of users because input and output dimensions are fixed. Furthermore, the number of trainable parameters is independent of the numbers of APs and users due to the reuse of the same information processing for all users and APs, realizing a high scalability and high generality to different numbers of users. With sufficient computational resources, the proposed GNN architecture can optimize a large-scale system.

⁵Note that J is the number of layers of NNs $\phi^{(l)}$ and $\gamma^{(l)}$ in layer l of the GNN. The GNN has L layers. In each layer, there are NNs $\phi^{(l)}$ and $\gamma^{(l)}$.

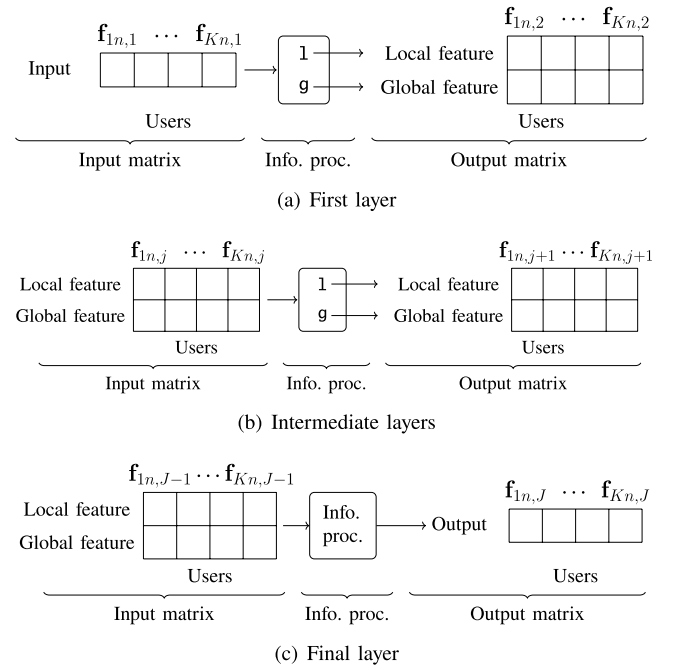


Fig. 4. Information processing (info. proc.) of $\gamma^{(l)}$ and $\phi^{(l)}$ of AP n . Note that this architecture is identical for all $l = 1, \dots, L$ and $n = 1, \dots, N$. The intermediate feature matrices contain local and global features defined in (7).

From (7) and (8), we can realize that the information processing is applicable to a varying number of users because the same calculations for local and global features are applied to all users. In this way, we meet the third requirement in Section II-B.

The inference described above is applied to both $\phi^{(l)}$ and $\gamma^{(l)}$. The message aggregation $\oplus$ is the aggregation of all incoming messages, which has the advantage that it allows for a variable number of neighbors.

C. Representation of Inference as Tensor Operations

Inference described in (6), (7) and (8) is intuitive to understand but not optimized for efficiency because it is done for each AP and user. If we apply this inference to large numbers of APs and users, we would need large nested loops, resulting in a very inefficient inference. However, it is possible to represent the inference for all APs and users as tensor operations, making parallel computing possible. This approach can significantly improve inference efficiency. We first describe a centralized inference representation with tensors for simulation-based training and testing, then show that the inference can be performed in a distributed way. In the centralized representation, input and output of a layer in $\phi^{(l)}$ and $\gamma^{(l)}$ are three-dimensional tensors, where the three dimensions are feature, AP and user, i.e., the input tensor $\mathbf{F}_l$ has a shape of $Q \times K \times N$ whereas the output tensor $\mathbf{F}_{l+1}$ has a shape of $2P \times K \times N$. The inference described in (7) is performed as

$$\mathbf{F}_{j+1} = \begin{pmatrix} \text{ReLU}(\mathbf{W}_j^1 \bullet \mathbf{F}_j + \mathbf{b}_j^1) \\ \mathbf{1} \cdot \text{ReLU}(\mathbf{W}_j^g \bullet \mathbf{F}_j + \mathbf{b}_j^g) / K \end{pmatrix}, \quad (9)$$

where $\mathbf{1}$ of shape $K \times K$ is a matrix of all ones, the multiplication of $\mathbf{1}$ and three-dimensional tensor $\mathbf{M}$ is the multiplication of $\mathbf{1}$ and the last two dimensions of $\mathbf{M}$, yielding a tensor where each row in dimension 2 is the sum of all rows in dimension 2 of $\mathbf{M}$, $\mathbf{A} \bullet \mathbf{B}$ is the multiplication of two-dimensional matrix $\mathbf{A}$ and the first dimension of three-dimensional tensor $\mathbf{B}$.

The tensor representation of (8) is similar to the first line of (9). Hence, the whole inference of $\phi^{(l)}$ and $\gamma^{(l)}$ is realized with tensors.

In order to represent message passing with tensors, let us define the adjacency matrix $\mathbf{E} \in \{0, 1\}^{N \times N}$, where element e_{mn} in row m and column n is 1 if there is an edge from node m to node n . Message passing and aggregation (the second step of (6)) can be expressed as

$$\mathbf{M} = \mathbf{F}_j^\phi \mathbf{E} / \sigma, \quad (10)$$

where $\mathbf{M} \in \mathbb{R}^{2P \times K \times N}$ is the aggregation of messages for all N APs in the third dimension, $\mathbf{F}_j^\phi \in \mathbb{R}^{2P \times K \times N}$ is the tensor of messages of all APs aggregated in the third dimension, σ is the vector of numbers neighbors of the APs. Similarly to before, the multiplication between $\mathbf{F}_j^\phi$ and $\mathbf{E}$ is done between the last two dimensions of $\mathbf{F}_j^\phi$ and $\mathbf{E}$. The division between $\mathbf{F}_j^\phi \mathbf{E}$ and σ is done elementwise in the second dimension of $\mathbf{F}_j^\phi \mathbf{E}$.

With tensor operation-enabled inference, efficient parallel computing can be applied for inference. Furthermore, the tensor operation unveils the symmetry in the inference, which is used to prove the HPE in Section IV-F.

Although the tensor representation described above is a centralized implementation on one computer, it is straightforward to modify it for distributed implementation. The N columns in the third dimension of $\mathbf{F}_l$ are stored in N APs. Since no operation in (9) requires information from multiple columns, (9) can be performed columnwise in a distributed manner, requiring no central unit for inference.

D. Iterative Inference

Since each AP can serve at most U users, each AP makes U decisions corresponding to U inferences of the GNN. In each inference, the input consists of the channel gain and the residual connection demand of each user. Initially, the residual demand is V for all users in the first inference, and it is updated after each decision for the next inference. After U inferences, the final assignment is the sum of all assignments in each iteration, and the residual demand is returned along with the assignment, which will be used as a penalty. The iterative inference process is illustrated in Figure 5.

E. Teacher-Student Model for Graph Pruning

By initialization, the graph is fully connected, i.e., every node is connected to every other node. Such a graph implies communication in the fronthaul even between APs far from each other, which is not necessary because it is very unlikely that they cooperate to serve the same users. However, manual graph pruning is difficult because there is no rule on the upper bound of distance that requires AP cooperation. Therefore, we

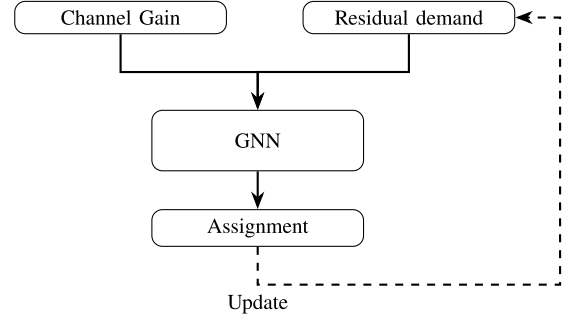


Fig. 5. The iterative inference of U times. In each inference, one user is assigned to each AP. Therefore, no more than U users are assigned to each AP. The residual demand to meet Constraint (1d) is updated after each iteration.

let GNN learn a sparse graph autonomously during training. For this purpose, we propose a teacher-student model inspired by [38]. Originally, this approach is used to transfer knowledge from large models (the *teacher*) to smaller, more efficient ones (the *student*), improving generalization and enabling deployment in resource-constrained settings. During training of the teacher model, an ℓ_1 -norm penalty is applied to encourage sparsity in the parameters. After training, parameters with magnitudes below a threshold are set to zero and treated as constants in the student model. This yields a smaller and more efficient student model that inherits the pruned structure from the teacher because the number of trainable parameters is smaller than the teacher model.

In this work, we focus on graph pruning. Unlike standard teacher-student models that prune NN parameters, we keep the same architectures for $\phi^{(l)}$ and $\gamma^{(l)}$ in both teacher and student models, but prune the graph by learning the adjacency matrix $\mathbf{E}$, because the message passing between APs is significantly more expensive and slower than local inference at each AP. Note that we focus on signaling overhead reduction via graph pruning, which has a subtle difference to fronthaul capacity. In a CF mMIMO network, message passing between two APs far away from each other is expensive and has a high latency due to the long distance and not very helpful, because two APs far from each other are unlikely to collaborate. Pruning the graph by removing edges between such edges can significantly reduce the signaling overhead, allowing for more data traffic given the same fronthaul capacity. Unlike previous works [9], [10] that model fronthaul capacity as a constraint, we focus on removing unnecessary edges in the graph without explicitly considering the fronthaul capacity. The teacher model is initialized with a fully connected graph. Moreover, we make a reasonable assumption that the distance between two APs is reciprocal to the necessity of cooperation. Define $\mathbf{D} \in \mathbb{R}^{N \times N}$ as the distance matrix, where d_{mn} ($m \neq n$) in row m and column n is the distance between AP m and AP n , the diagonal element d_{nn} in row n and column n is set as a large number (e.g., 10^6 , since an AP does not pass message to itself, as will become clear in (11)), the adjacency matrix $\mathbf{E}$ is computed as

$$\mathbf{E} = \mathbf{1} - \psi \left(\frac{\mathbf{D}}{\beta} - c \right), \quad (11)$$

where $\mathbf{1}$ of shape $N \times N$ is a matrix of all ones, $\psi(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function, β is a tunable parameter, and c is a constant. Note that (11) is only one possibility for characterizing the relationship between edge weights and distances between APs. Other functions can also be used as long as they reflect the relationship between necessity of collaboration and geographical distance. Values of β and c are initialized such that the off-diagonal elements of $\mathbf{E}$ are close to 1 (i.e., a fully connected graph) at the beginning of the teacher model training. After the training has converged, β reduces linearly until it reaches a very small value such that $\mathbf{E}$ becomes a matrix of all zeros (i.e., no edges in the graph), cooperation between APs is no longer possible and the performance degrades. With the training result of the teacher model, we select a value of β before the performance begins to degrade significantly, compute the corresponding matrix $\mathbf{E}$, and round the elements to 0 or 1 for the student model. The student model is then trained with the fixed, sparse matrix $\mathbf{E}$. If properly trained, the student model is expected to achieve a similar performance to the teacher model, as we will see in Section VI.

Remark 2: The GNN architecture allows for arbitrary numbers of APs. The specific structures of $\phi^{(l)}$ and $\gamma^{(l)}$ also allow for arbitrary numbers of users. However, the adjacency matrix $\mathbf{E}$ with a size of $N \times N$ is only applicable to a specific number of APs. The pruned graph is learned for a specific network deployment. Since the GNN training is much cheaper than network deployment, it is practical to train a new GNN optimized for a certain network deployment rather than pursuing a zero-shot generalization to different networks with one trained GNN.

F. Hierarchical Permutation-Equivariance

While permutation-equivariance (PE) is a well-known property of GNNs, it only guarantees that the permutation of nodes (i.e., APs) leads to the same permutation in the output. However, as described in Section II, we have two types of entities: APs and users. Both types can be permuted independently, and the output assignment matrix should be permuted accordingly. This leads us to the concept of HPE, which guarantees that permutations of APs and users of each AP lead to the same permutations in the output.

If we can realize HPE, a high generalizability can be achieved because it guarantees the same performance for data samples that do not exist in the training set $\mathcal{D}$ but can be obtained by permuting a data sample in $\mathcal{D}$. Although HPE does not exist in conventional NN architectures, the proposed customized GNN is inherently HPE.⁶

To describe a permutation, we define a square permutation matrix $\mathbf{P}$, where there is only one “1” in each row and each column. For example, define a permutation matrix

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (12)$$

⁶When a model is said to be HPE, we mean that it satisfies this property.

$\mathbf{P}\mathbf{G}$ permutes the second and third rows of $\mathbf{G}$, whereas $\mathbf{G}\mathbf{P}^T$ permutes the second and third columns of $\mathbf{G}$. It is obvious that

$$\mathbf{P}^T\mathbf{P} = \mathbf{I} \quad (13)$$

for any permutation matrix $\mathbf{P}$ because $\mathbf{P}$ and $\mathbf{P}^T$ are inverse operations to each other.

Let $\mathbf{P}_a \in \{0, 1\}^{N \times N}$ be a permutation matrix for APs and $\mathbf{P}_u \in \{0, 1\}^{K \times K}$ be a permutation matrix for users. We further assume $\mathbf{S} = N_\theta(\mathbf{G})$. We define that N_θ is HPE if

$$\mathbf{P}_u\mathbf{S}\mathbf{P}_a^T = N_\theta(\mathbf{P}_u\mathbf{G}\mathbf{P}_a^T) \quad (14)$$

for any $\mathbf{P}_u$ and $\mathbf{P}_a$.

Theorem 1: The proposed GNN architecture is HPE.

Proof: The proof can be found in Appendix. $\square$

V. TRAINING FOR CONSTRAINED COMBINATORIAL PROBLEM WITH AUGMENTED LAGRANGIAN METHOD

In Section IV, we proposed a customized GNN that reflects problem properties 2)–5) in Section II-B. The remaining property 1) (combinatorial nature) is addressed in this section. The binary optimization variable $\mathbf{S}$ makes Problem (1) nondifferentiable. As a result, conventional NN training methods based on gradient ascend cannot be applied. Although there are approaches for combinatorial problems with ML [40], [41], [42], they are severely limited to meet the requirements described in Section II-B, see Section I for details. Therefore, we propose a novel relaxation technique that relaxes the optimization variable $\mathbf{S}$ from binary to continuous variables between 0 and 1, i.e., $\mathbf{S} \in [0, 1]^{K \times N}$, and introduce an additional constraint inspired by the information entropy:

$$-\sum_{k=1}^K \sum_{n=1}^N s_{kn} \log(s_{kn}) = 0. \quad (15)$$

The information entropy is nonnegative and it is 0 if and only if $s_{kn} = 0$ or $s_{kn} = 1$. Therefore, Constraint (15) guarantees that $s_{kn} = 0$ or 1 for any k and n . We apply ALM for this constrained optimization problem. The basic idea of ALM is to address constraints as linear and quadratic penalty terms, and to adapt weights of penalty terms for efficient convergence. In the relaxed problem, Constraint (1c) is guaranteed by letting N_θ make U decisions, each selecting one user (see Section IV-D). Constraints (1d) and (15) are addressed as penalties. The objective function is therefore formulated as

$$g = f - \lambda_1 \sum_{k=1}^K \max \left\{ V - \sum_{n=1}^N s_{kn}, 0 \right\} - \frac{1}{2} \lambda_2 H - \nu_1 \sum_{k=1}^K \max \left\{ V - \sum_{n=1}^N s_{kn}, 0 \right\}^2 - \frac{1}{2} \nu_2 H^2, \quad (16)$$

where λ_j and ν_j ($j = 1, 2$) are the weights of the linear and quadratic penalty terms, respectively, and $H = -\sum_{k=1}^K \sum_{n=1}^N s_{kn} \log(s_{kn})$ is the penalty term defined in (15) (note that $H \geq 0$ and the equality holds if and only if $s_{kn} = 0$ or 1).

TABLE I
IMPORTANT SCENARIO PARAMETERS

Parameter	Value
Number of APs	5 (small network scenario), 56 (large network scenario)
Number of users	2–4 (small network scenario), 25–40 (large network scenario)
Samples in training set	10 240
Samples in testing set	1024
Minimum number of serving APs per user L	2
Maximum number of served users per AP U	2
Number of layers in γ	3
Number of layers in ϕ	2
Number of GNN layers	5
Feature dimension	5
Optimizer	Adam with learning rate adaptation
Initial learning rate	1.7×10^{-4}
Minimum learning rate	1×10^{-5}
Batch size	512

This proposed approach is a novel combination of analytical ALM and ML. Grounded in the Karush–Kuhn–Tucker (KKT) conditions, ALM is a principled method for solving constrained optimization problems with good robustness and reproducibility. Unlike conventional ALM, which optimizes $\mathbf{S}$ directly, ALM in the proposed approach is used to formulate the training objective (16) and $\mathbf{S}$ is optimized by updating the GNN parameter θ using an NN optimizer (e.g., ADAM). This combined approach of analytical optimization and ML can be generalized to other analytical gradient-based algorithms.

At the beginning of training, λ_1 , λ_2 , ν_1 and ν_2 are initialized as 0. After the objective stops to increase, λ_1 and ν_1 are updated according to the ALM rule. The training continues with the new objective function. This alternating updating scheme of penalty weights and GNN parameters repeats until Constraint (1d) is satisfied. The same update rule is then applied to λ_2 and ν_2 until Constraint (15) is satisfied. The trained GNN can now compute valid solutions to Problem (1).

After convergence with the satisfied constraints, for the teacher model, β is updated as described in Section IV-E to determine a sparse matrix $\mathbf{E}$. For the student model, β is chosen and the smallest value without considerable performance degradation. This value is kept constant throughout the training.

Compared to listing all combinations of assignments of all APs [40], the proposed approach is more scalable because the number of assignment combinations grows exponentially with the number of APs. As opposed to rounding in testing [41] (rounding is impossible in testing because it is nondifferentiable), the proposed approach is consistent in training and testing and can fulfill the constraints better in testing with unseen data. Compared to relaxation with the CONCRETE distribution with the temperature as a hyperparameter [42], which is either a constant or is decreasing in a constant speed, hyperparameters of the proposed method (λ_1 , λ_2 , ν_1 , and ν_2) are regulated by the ALM rules based on the feedback of previous training results. In our experiment, our

Algorithm 1 Training Using ALM for Teacher and Student Models

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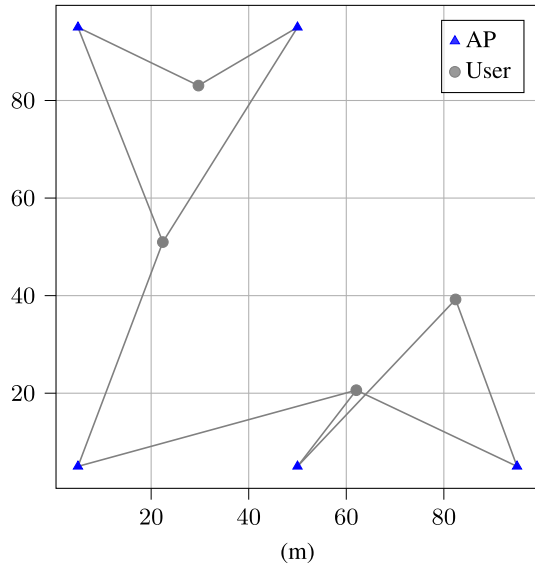
1: if teacher then
2:    $\beta = 1000$ 
3: else
4:   Set  $\beta$  as the smallest value without considerable performance loss
5:   Prune the graph according to the edge weights
6: end if
7: Set  $\lambda_j = \nu_j = 0$  for  $j = 1, 2$ .
8: repeat
9:   Update GNN with a gradient ascent step.
10: until Convergence
11: repeat
12:    $\lambda_1 \leftarrow \lambda_1 + \nu_1 \sum_{k=1}^K \text{ReLU} \left( L - \sum_{n=1}^N s_{kn} \right)$ 
13:    $\nu_1 \leftarrow \nu_1 + \Delta \nu$ 
14: repeat
15:   Update GNN with a gradient ascent step.
16: until Convergence
17: until  $\sum_{k=1}^K \text{ReLU} \left( L - \sum_{n=1}^N s_{kn} \right) = 0$ .
18: repeat
19:    $\lambda_2 \leftarrow \lambda_2 + \nu_2 \sum_{k=1}^K \text{ReLU} \left( L - \sum_{n=1}^N s_{kn} \right)$ 
20:    $\nu_2 \leftarrow \nu_2 + \Delta \nu$ 
21: repeat
22:   Update GNN with a gradient ascent step.
23: until Convergence
24: until  $-\sum_{k=1}^K s_{kn} \log(s_{kn}) = 0$ .
25: if teacher then
26:   repeat
27:      $\beta = \beta - \Delta \beta$ 
28:     Update GNN with a gradient ascent step.
29:   until  $\beta = 0$ 
30: end if

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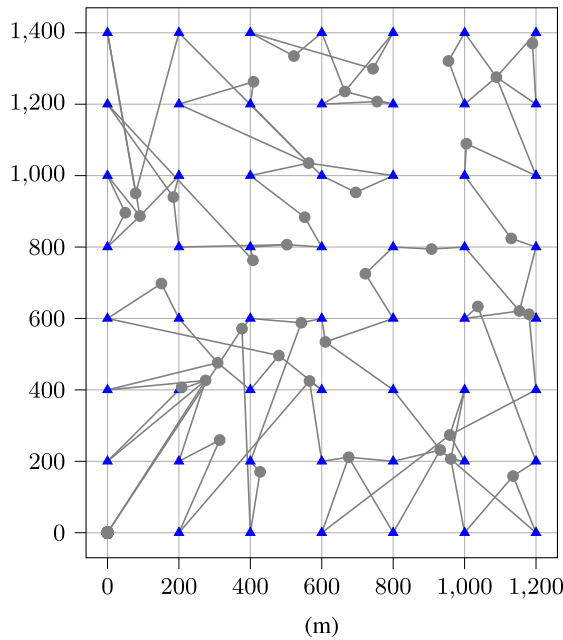
proposed method can realize a better performance subject to the constraints.

Like conventional NN training, the big training set $\mathcal{D}$ is divided into multiple batches. However, unlike conventional NN training, where the NN is typically designed for a fixed input size, we require that the proposed GNN is capable of serving a varying number of users. Since each batch input is represented as a tensor, the number of users constitutes one of its dimensions, and thus must be identical across all samples within a batch to ensure tensor compatibility during training. Therefore, the total training data set is generated with the maximum intended number of users. For each batch, a subset of data samples is randomly selected, and for every sample within the batch, a random user count is drawn. Subsequently, the corresponding inputs are trimmed to match this user count, effectively allowing the network to observe and adapt to different user scenarios across batches. This approach enables the GNN to generalize across a range of user numbers while remaining compatible with mini-batch training requirements.

Summarizing the description above, the training process is formulated as Algorithm 1.



(a) Small network scenario with 5 APs



(b) Large network scenario with 56 APs

Fig. 6. Visualization of AP-user assignment in two representative scenarios.

TABLE II
NUMBERS OF EDGES IN DIFFERENT MODELS

Model	Number of edges
Teacher (fully connected)	3080
Good student	362 (12% of fully connected graph)
Bad student	194 (6% of fully connected graph)

VI. TRAINING AND TESTING RESULTS

Training and testing results using the proposed method are presented in this section. We consider a small network scenario with 5 APs and 2–4 users, where the performance upper bound can be found by exhaustive search, and a large

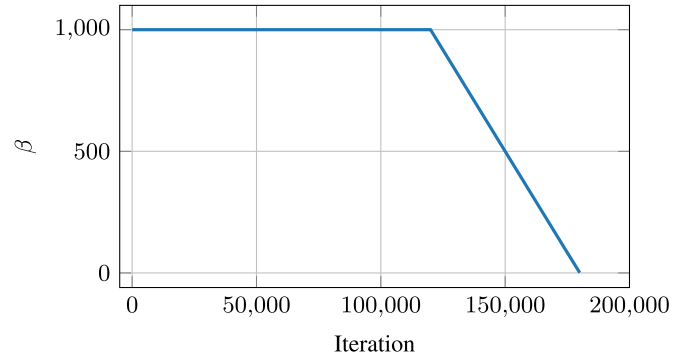
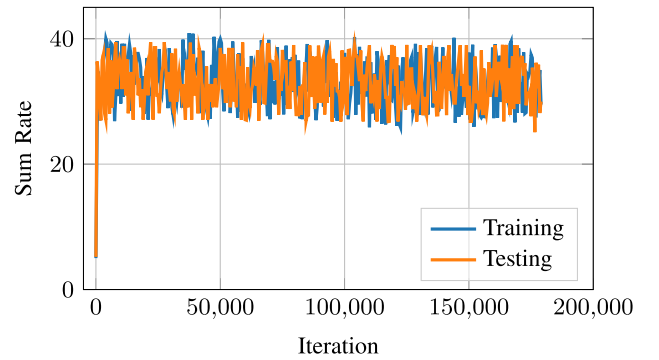
Fig. 7. The value of β during the teacher model training. The value is constant before convergence, and then decreases linearly.

Fig. 8. Realized sum rates in training and testing with the fully connected graph (teacher). Note that the varying sum rate is due to the random user numbers (see Section II-B), not poor convergence.

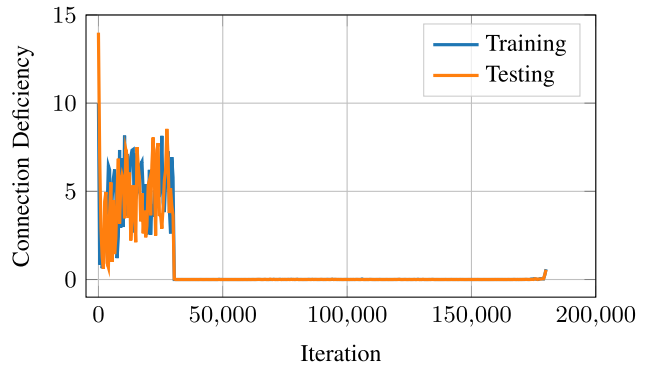


Fig. 9. Connection deficiency penalties in training and testing with the fully connected graph (teacher).

network scenario with 56 APs and 25–40 users, where the scalability of the proposed method is evaluated. The two scenarios are illustrated in Figure 6. Important scenario and GNN parameters are shown in Table I.

As described in Section V, the training maximizes f first, then adds constraints as penalties in the objective function one by one. For the teacher model, β decreases to optimize the trade-off between graph sparsity and performance, as shown in Figure 7. Applying the above setting, Figure 8 shows the sum rate achieved in training and testing of the teacher model. We can observe that the sum rate is improved quickly in the beginning of training, then degrades slightly due to the introduction of constraints. The fluctuation is mainly due to the

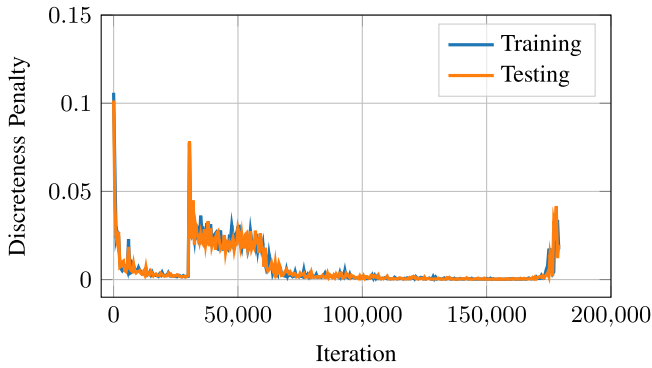


Fig. 10. Discreteness penalties in training and testing with the fully connected graph (teacher).

TABLE III
ACHIEVED SUM RATE (BIT/HZ/S) OF PROPOSED METHOD
AND BASELINES

	Small network scenario	Large network scenario
Teacher	1.14	33.87
Good student	—	33.52
Bad student	—	32.17 (Constraint (1d) not always satisfied)
Upper bound	1.15	Not available
Random	0.60	19.42
Simulated annealing	1.08	33.04
Genetic algorithm	1.05	32.72
GSD	1.06	26.14
μ_R	1.07	28.41

varying number of users per batch (see Section V). Moreover, the decreasing β does not have an observable impact on the sum rate. The main reason is that the sum rate is roughly optimized if every AP serves its U closest users. This naive strategy does not require AP collaboration. We will see later that the absence of AP cooperation mainly impacts Constraint (1d). Finally, the training and testing curves are very similar, suggesting good generalizability of the trained model.

Figure 9 shows the connection penalty (i.e., the penalty that a user is not served by at least V APs) in training and testing of the teacher model. It is clear that the connection penalty is high before it is taken into the objective function. After the penalty term is introduced after about 3×10^4 steps, the connection penalty is reduced to 0 and Constraint (1d) is fulfilled. However, at the end of the training, the connection penalty increases again as β decreases close to 0 and hence AP collaboration becomes increasingly difficult, suggesting that APs need to collaborate to ensure that each user is served by at least L APs, which matches our intuition. Furthermore, we observe again that the training and testing curves are similar to a large extent, implying a good generalizability of the trained model to unseen data.

After the minimum connection number is ensured, the penalty for discreteness is introduced to enforce that the optimization variables are either 0 or 1. The discreteness penalty in training and testing of the teacher model is shown in Figure 10. We observe that the discreteness penalty is high at the beginning of training, reduces to almost 0 after the

introduction of the penalty terms until β reduces to a small value. Similarly to the connection penalty, the overlapping training and testing curves confirm the generalizability of the trained model.

In the following, we demonstrate the graph pruning in the big network scenario with 56 APs, which reveals the relationship between number of edges and performance more clearly than the small network scenario. Based on the training results of the teacher model, we define two student models: a “good student” model with $\beta = 65$ and a “bad student” model with $\beta = 40$. The fully connected and pruned graphs are shown in Figure 11. Numbers of edges in each model are listed in Table II. Because the good student model has only 12% of the edges of the fully connected graph, and the computational effort to compute the messages in the GNN is proportional to the number of edges, the good student model requires only 12% of the computational effort of the fully connected graph. Furthermore, the pruned graph has not only fewer, but also shorter edges than the fully connected graph. In the large network scenario, the longest edge in the pruned graph is 15% of the longest edge in the fully connected graph, realizing a significant reduction in communication latency because the GNN inference is done in a synchronized way, i.e., a node needs to wait for the messages from all its neighbors before node processing. Therefore, the inference time is determined by the longest edge in the graph.

The training and testing curves of the two student models are shown in Figure 12. While the good student has a similar training and testing behavior as the teacher model, the bad student model training realizes a worse sum rate. This observation shows the trade-off between performance and graph sparsity.

Similar observations can be found in the penalties for connection numbers and discreteness, as shown in Figure 13 and Figure 14, respectively. While the good student model has a similar result to the teacher model, the bad student model is deficient in ensuring the constraints. The difference between good and bad student models in fulfilling the minimum connection requirement is more significant than the sum rate. As explained above, the sum rate can be maximized by the naive tactic that every AP serves the closest U users without AP collaboration. On the contrary, the minimum connection requirement can only be retained with AP collaboration. To realize the above performance, a training complexity of 55 peta float-point operations (PFLOPs) (55×10^{15} float-point operations (FLOPs)) is required. Compared to this level of complexity, training the typical “heavy” ResNet-50 [50] on ImageNet dataset has a complexity of 1.4 EFLOPs (1.4×10^{18} FLOPs). Training the light-weight EfficientNet-B0 [51] on ImageNet dataset has a complexity of about 176 PFLOPs. Both are significantly higher than the proposed approach. Since training is done offline and only needs to be done once for a given network deployment, the training cost is significantly less than network deployment cost, and is not a hurdle for future CF mMIMO networks.

Compared to off-line training, real-time inference is more sensitive to computational complexity. In our experiment, the inference complexity is about 0.40 GFLOPs in the large network scenario. As a comparison, EfficientNet-B0 has an

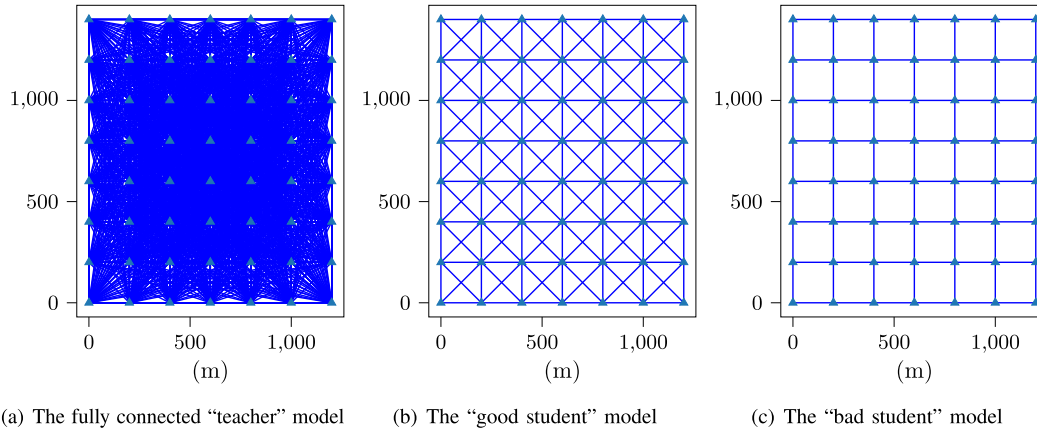


Fig. 11. Fronthaul connections of the three models.

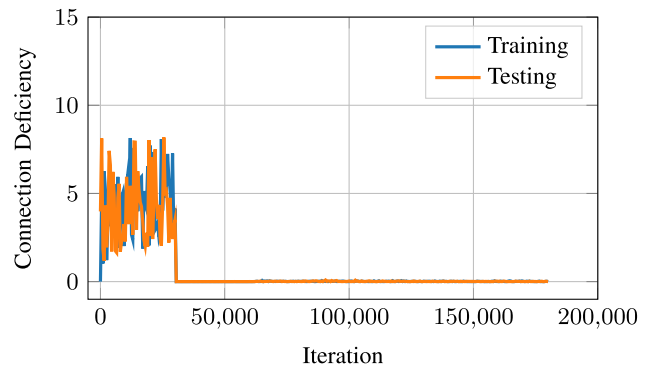
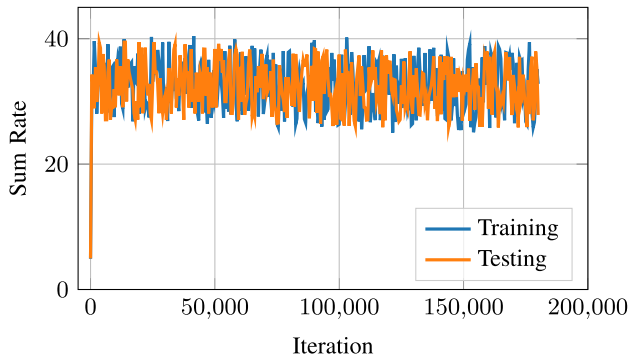
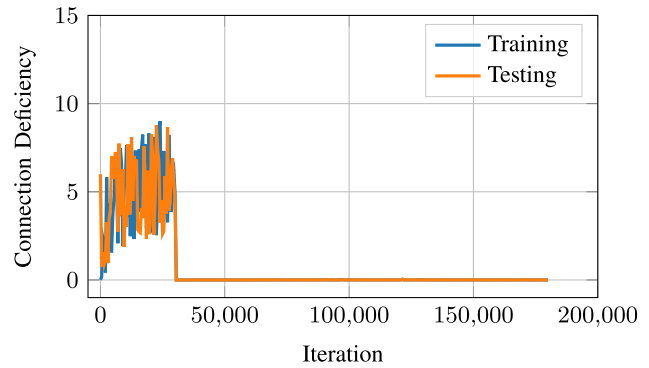
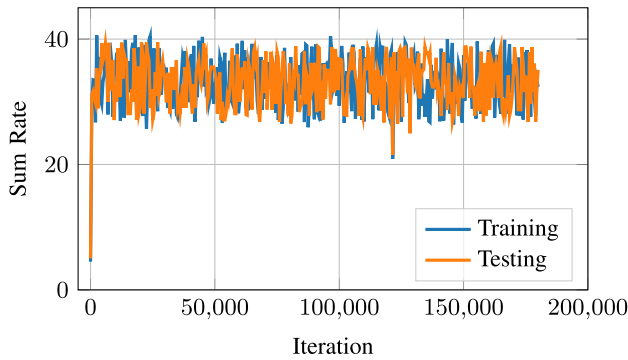


Fig. 12. Realized sum rates in training and testing with the pruned graph (student).

Fig. 13. Connection deficiency penalties in training and testing with the pruned graph (student).

inference complexity of about 0.39 GFLOPs per image. ResNet-50 has an inference complexity of about 3.8 GFLOPs per image. We can see that the inference complexity of our proposed GNN is among the lowest of existing deep learning models proposed several years ago. It will not be a burden for future CF mMIMO networks with powerful edge servers.

Table III shows the comparison between the proposed approach, an upper bound found by exhaustive search, and baseline algorithms. Note that the graph pruning only achieves reasonable results in the large network scenario because all 5 APs are "neighbors" to each other in the small network

scenario. Moreover, the upper bound is found by exhaustive search through $\binom{K}{U}^N$ possible assignments. In the small network scenario, there are 7776 possible assignments. In the large network scenario, this number is 9×10^{161} , which is not feasible for exhaustive search. We can conclude that the teacher model is very close to the upper bound in the small network scenario and outperforms all the baselines. The good student model is close to the teacher model in the large network scenario and still outperforms the baselines. The bad student model is worse than the teacher model in sum rate and cannot fulfill the constraints consistently. Finally, we note

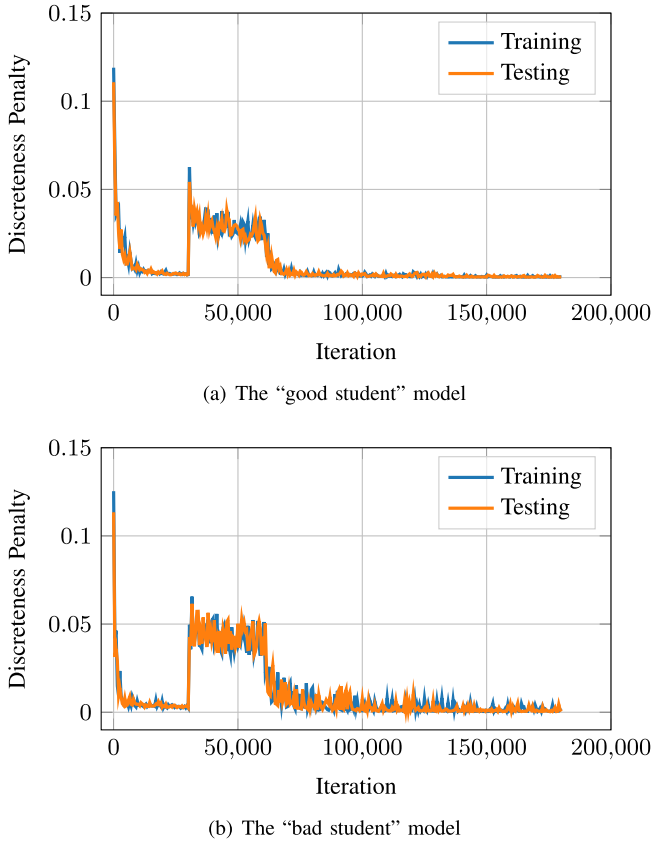


Fig. 14. Discreteness penalties in training and testing with the pruned graph (student model).

that baseline algorithms simulated annealing (SA) [52], genetic algorithm (GA) [53], generalized serial dictatorship (GSD) [54] and μ_R [55] are centralized algorithms and require more computation time than the proposed method. In addition, GA and SA require significant more time to converge. The student model outperforms them in a distributed way on a sparse graph with significantly less computation time.

VII. CONCLUSION

millimeter wave (mmWave) CF mMIMO is a promising solution to next-generation wireless communications. On the one hand, mmWave band provides large bandwidths to realize high data rates. On the other hand, cooperative APs in CF mMIMO are a countermeasure against high path loss and blockage loss. Due to channel and semiconductor properties, mmWave CF mMIMO optimization is largely an assignment problem between APs and users. Due to the high scalability requirement (many APs and users) and the combinatorial nature of the assignment problem, we do not have a satisfactory solution in the literature. In this paper, we have proposed an unsupervised ML approach, which looks for the optimal solution autonomously without labels. The CF mMIMO network is described as a graph, where nodes represent APs and edges between nodes are fronthaul and enable cooperation between APs. A highly customized GNN has been proposed, which assigns a fixed number of APs to a varying number of users. The GNN has the property of HPE, ensuring generalizability of the trained model with

permuted input data. Moreover, we have learned a sparse graph for GNN inference, which has much lower communication overhead in fronthaul compared to the fully connected graph. We have shown that with a carefully pruned graph, the GNN achieves a similar performance with significantly less edges. Finally, we have addressed the difficulty that the combinatorial assignment optimization problem has discrete variables and is therefore non-differentiable. We relaxed the problem to a continuous one, introduced an entropy-inspired constraint to ensure that the relaxed problem is equivalent to the original problem, and combined ALM and GNN training to solve the constrained optimization problem. Simulation results showed that the proposed solution has a high generalizability to unseen data, and is advantageous compared to baseline algorithms in performance, computation time, and distributed inference with reduced signaling in the fronthaul.

This work can be extended to other optimization objectives in CF mMIMO. The graph attention network can be introduced to dynamically determine which edges are “activated” according to the current network state.

Code and data sets are available under https://github.com/bilepeng/cfmmimo_user_assignment.

APPENDIX PROOF OF THEOREM 1

We first prove that a layer of ϕ and γ is HPE, then show that ϕ and γ as a whole is HPE. Combined with the fact that message passing and aggregation is also HPE, we can prove that the proposed GNN is HPE.

When the layer input $\mathbf{F}_i$ is permuted as $\mathbf{P}_u \mathbf{F}_i \mathbf{P}_a^T$, the inference of a layer (9) becomes

$$\begin{aligned}
 & \begin{pmatrix} \text{ReLU}(\mathbf{W}_i^1 \bullet \mathbf{P}_u \mathbf{F}_i \mathbf{P}_a^T + \mathbf{b}_i^1) \\ \mathbf{1} \text{ReLU}(\mathbf{W}_i^g \bullet \mathbf{P}_u \mathbf{F}_i \mathbf{P}_a^T + \mathbf{b}_i^g)/K \end{pmatrix} \\
 &= \begin{pmatrix} \mathbf{P}_u \text{ReLU}(\mathbf{W}_i^1 \bullet \mathbf{F}_i \mathbf{P}_a^T + \mathbf{b}_i^1) \\ \mathbf{1} \mathbf{P}_u \text{ReLU}(\mathbf{W}_i^g \bullet \mathbf{F}_i \mathbf{P}_a^T + \mathbf{b}_i^g)/K \end{pmatrix} \\
 &= \begin{pmatrix} \mathbf{P}_u \text{ReLU}(\mathbf{W}_i^1 \bullet \mathbf{F}_i \mathbf{P}_a^T + \mathbf{b}_i^1) \\ \mathbf{P}_u \mathbf{1} \text{ReLU}(\mathbf{W}_i^g \bullet \mathbf{F}_i \mathbf{P}_a^T + \mathbf{b}_i^g)/K \end{pmatrix} \\
 &= \begin{pmatrix} \mathbf{P}_u \text{ReLU}(\mathbf{W}_i^1 \bullet \mathbf{F}_i + \mathbf{b}_i^1) \mathbf{P}_a^T \\ \mathbf{P}_u \mathbf{1} \text{ReLU}(\mathbf{W}_i^g \bullet \mathbf{F}_i + \mathbf{b}_i^g)/K \mathbf{P}_a^T \end{pmatrix} \\
 &= \mathbf{P}_u \mathbf{F}_{i+1} \mathbf{P}_a^T. \tag{17}
 \end{aligned}$$

The second line of (17) is valid because multiplication $\mathbf{W}_i \bullet \mathbf{F}_i$ is done between $\mathbf{W}_i$ and the first dimension of $\mathbf{F}_i$ (we ignore the superscripts 1 and g because they follow the same logic), while multiplication $\mathbf{P}_u \mathbf{F}_i$ is between $\mathbf{P}_u$ and the last two dimensions of $\mathbf{F}_i$. As a result, the multiplication is commutative. The third line of (17) is true because matrix multiplication is associative and according to the definition of $\mathbf{P}_u$, there is only one “1” in every row and every column. Therefore, $\mathbf{P}_u \mathbf{1} = \mathbf{1} \mathbf{P}_u = \mathbf{1}$. The fourth line of (17) is valid because the addition between $\mathbf{W}_i \bullet \mathbf{F}_i$ and $\mathbf{b}_i$ is in the first dimension and the multiplication between $\mathbf{F}_i$ and $\mathbf{P}_a$ is in the second and third dimensions. Therefore, $\mathbf{P}_a$ has no impact on $\mathbf{b}_i$. The fifth line is obtained when we plug (9) into the fourth line of (17). In this way, we prove that a layer of ϕ and γ is HPE. Since every layer is HPE, the whole NNs ϕ and γ are also HPE.

Next, we prove that the message passing and aggregation are HPE. When APs are permuted by $\mathbf{P}_a$, the adjacency matrix $\mathbf{E}$ becomes $\mathbf{P}_a \mathbf{E} \mathbf{P}_a^T$. Applying the property (13), message passing and aggregation (10) turns into

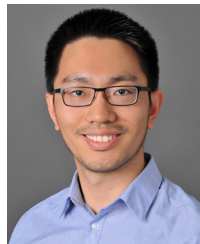
$$\mathbf{P}_u \mathbf{F}_I^\phi \mathbf{P}_a^T \mathbf{P}_a \mathbf{E} \mathbf{P}_a^T / \sigma = \mathbf{P}_u \mathbf{F}_I^\phi \mathbf{E} \mathbf{P}_a^T / \sigma. \quad (18)$$

As all ϕ , γ , message passing and aggregation are HPE, the proposed customized GNN is HPE.

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